A FINITE EXTENSION IS ALGEBRAIC, AND ALGEBRAICITY IS DETECTED BY FINITENESS OF ALL MONOGENIC SUBEXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Between statement and proof the author inserts the standalone remark 'In general, it is very false that an algebraic extension is finite' (source lines 884-885); it is author commentary outside both counted spans and is not delivered.

Editorial note. The label lemma-finite-is-algebraic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter examples: example-degree-rational-function-field and example-degree-simple-algebraic-extension (a monogenic extension k(alpha)/k is finite exactly when alpha is algebraic over k)

Source reference. Original reference 09G6: sources/stacks-fields/pinned-original-source.tex, source lines 683–683. Prerequisite pointer invoked at 896; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09G7: sources/stacks-fields/pinned-original-source.tex, source lines 721–721. Prerequisite pointer invoked at 897; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *A finite extension is algebraic. In fact, an extension E/k is algebraic if and only if every subextension $k(\alpha)/k$ generated by some $\alpha \in E$ is finite.*

Proof. Let E/k be finite, say of degree n. Choose $\alpha \in E$. Then the elements $1, \alpha, \ldots, \alpha^n$ are linearly dependent over k, or we would necessarily have $[E : k] > n$. A relation of linear dependence now gives the desired polynomial that α must satisfy.

For the last assertion, note that a monogenic extension $k(\alpha)/k$ is finite if and only if α is algebraic over k, by Examples 09G6 and 09G7. So if E/k is algebraic, then each $k(\alpha)/k$, $\alpha \in E$, is a finite extension, and conversely. □

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